7.8 Optimizing performance

By default, S3 handles 3,500 writes and 5,500 reads per second. If your workload requires higher throughput, you need to consider the following when coming up with the naming scheme for the object keys.

Objects are stored without a hierarchy on S3. There is no such thing as a directory. All you do is specify an object key, as discussed at the beginning of the chapter. However, using a prefix allows you to structure the object keys.

By default, the slash character (`/`) is used as the prefix delimiter. So, `archive` is the prefix in the following example of object keys:

```
archive/image1.png
archive/image2.png
archive/image3.png
archive/image4.png
```

Be aware that the maximum throughput per partitioned prefix is 3,500 writes and 5,500 reads per second. Therefore, you cannot read more than 5,500 objects from the prefix `archive` per second.

To increase the maximum throughput, you need to distribute your objects among additional prefixes. For example, you could organize the objects from the previous example like this:

```
archive/2021/image1.png
archive/2021/image2.png
archive/2022/image3.png
archive/2022/image4.png
```

By doing so, you can double the maximum throughput when reading from archive/2021 and archive/2022 as illustrated in figure 7.6.